Austin Moon

(647) 523-9411 a8moon@uwaterloo.ca linkedin.com/austinmoon github.com/Aust1n-moon website

Education

University of Waterloo Waterloo, ON

B.A.Sc. Mechatronics Engineering, Specialization in Artificial Intelligence *Sep. 2024 – Expected June 2029*

Technical Skills

Languages: Python, C/C++, SQL, VHDL
AI/ML & Vision: PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, ByteTrack, NumPy, pandas
Embedded/Tools: STM32, Arduino, PLC, I2C/SPI/UART, CAN bus, Git, Linux, Docker
Design/Hardware: SolidWorks, AutoCAD, Altium, KiCad, PCB Fabrication, Soldering, Rapid Prototyping, Excel

Experience

Machine Learning Engineer Sep 2026 – Present

WAT.ai – OffBall XI *Waterloo, ON*

- Working on reinforcement learning (Google Research Football) and graph-based models (PyTorch Geometric) to simulate counterfactual off-ball movement and decision-making from professional soccer tracking data
- Ran Google Research Football simulations to model a player's movement and presented findings to the team

Tech Executive – AI/ML, Full-Stack Engineer Sep 2026 – Present

UW Association of Korean-Canadian Scientists and Engineers *Waterloo, ON*

- Building an AI resume tailoring app that classifies job postings and scores resume bullets with Jev, rewrites them with LLM APIs using field specific prompts, and generates tailored resumes

Embedded Software Developer May 2025 – Aug 2026

Waterloo Aerial Robotics Group *Waterloo, ON*

- Developed STM32 firmware in C to drive a servo motor with PWM from SPI-based ADC input in real time
- Developed a C++ driver for a three-axis magnetometer on STM32, using I2C communication with DMA to read magnetic flux density and compute a real-time compass heading
- Extracted and parsed position, altitude, and speed from a u-blox M10 GPS receiver through an interrupt-driven UART NMEA stream, converting the data into navigation coordinates
- Implemented a low-pass filter in C for the ICM-42688 IMU within WARG's firmware, reducing noise in sensor data

Mechatronics Engineering Intern Sep 2025 – Dec 2025

Spyder Controls Corporation *Lacombe, AB*

- Led a UV chamber rebuild, wiring the control box, LED chamber, and cooling system with relays and sensors
- Built and repaired 15+ test jigs, diagnosing and fixing faulty circuitry and mechanical failures
- Flashed and validated firmware and circuits on 10+ products, systems, and test jigs over a CAN bus network
- Wrote a Python script for an Arduino Nano to flag out-of-tolerance (±5%), open, and shorted thermistors

Mechanical Engineer Feb 2025 – Nov 2025

UW Orbital *Waterloo, ON*

- Designed a modular 3-axis ADCS test platform in SolidWorks, gimballing ±30° in x/y with free z-axis rotation
- Engineered a custom bearing-supported gimbal on all 3 axes for low-friction, dynamic CubeSat orientation testing
- Implemented a stabilizing counterweight and reinforcement structure to support a minimum 4 kg load

Engineering Assistant Jan 2025 – Apr 2025

University of Waterloo *Waterloo, ON*

- Supervised the Rapid Prototyping Centre and Engineering Student Machine Shop, supporting 40+ students per day
- Set up and operated a new Rayjet R500 laser cutter, reviewing technical documentation and troubleshooting issues
- Processed 200+ laser cutting, 150+ 3D printing, and 50+ water jetting jobs using SolidWorks and AutoCAD
- Designed 2 drone attachments in SolidWorks in collaboration with the Waterloo Regional Police ERT

Projects

NFL Defensive Coverage Predictor | *Python, PyTorch, YOLOv11, ByteTrack, OpenCV* | *GitHub*

- Trained a YOLOv11 detector on 10,000+ images (69% mAP50, 75% precision) with ByteTrack multi-object tracking
- Built automated camera calibration with Hough line transforms to map image pixels to real-world coordinates
- Engineered 339-dimensional features from 4.8M rows of tracking data to train a PyTorch MLP on 6 coverage types

G-Helper Clone | *Python, PyQt6, D-Bus, Linux* | *GitHub*

- Developed a PyQt6 system tray app to manage ASUS laptop hardware on Fedora Linux using Claude AI
- Tuned 50+ kernel and hardware parameters (CPU EPP, PCIe ASPM, PCI runtime PM), reducing idle power by 80%
- Built a debounced state machine to auto-configure CPU frequency, GPU DPM, and fan PWM on battery power